Tohar Hermon | Computer Engineer

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PROFESSIONAL SUMMARY : Computer Engineering graduate with a strong academic background and hands-on experience in full-stack development, cloud architecture, and systems programming. Proven ability to design and implement end-to-end solutions, from low-level compilers to scalable cloud-based CRM systems. Fast learner with a "can-do" attitude, experienced in leading teams and managing complex logistics under high-pressure environments.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, C#, PHP, SQL, JavaScript, Java, MATLAB, Verilog.
- **Cloud & Backend:** AWS (Amplify, DynamoDB, Cognito, Lambda, S3), Node.js, Next.js, Vercel.
- **Tools & Systems:** Git, Docker, Linux, RISC-V Architecture, Flex & Bison, OpenCV, FPGA.
- **Core Competencies:** OOP, Design Patterns, Serverless Architecture, Embedded Systems, Computer Vision.

PROJECTS:

CPL to RISC-V Compiler | C, Lex, Bison, RISC-V

- Developed a full-scale compiler translating a custom language (CPL) into RISC-V Assembly.
- Implemented all compilation stages: Lexical Analysis (Flex), Syntax Parsing (Bison), and Semantic Analysis with complex symbol table management.
- Engineered a code generation engine optimized for low-level RISC-V architecture constraints.

Cloud-Based CRM System | Next.js, AWS, Tailwind CSS

- Architected a serverless enterprise solution using AWS (Amplify, DynamoDB, Cognito) for scalable data management and secure authentication.
- Developed a dynamic dashboard for real-time task scheduling and client analytics.
- Deployed a robust CI/CD pipeline using GitHub and Vercel for rapid iteration.

Smart Door Lock System | Python, OpenCV, Raspberry Pi

- Integrated a real-time biometric security solution featuring face recognition and RFID multi-factor authentication.
- Optimized image processing pipelines to ensure low-latency performance on edge hardware.
- Designed secure user management logs and local SQL database integration.

Sign Language Translation Tool | Python, Machine Learning

- Developed a real-time system translating ASL gestures into text using a QDA classifier.
- Designed to enhance communication accessibility for the hearing impaired.

WORK EXPERIENCE: Security Shift Supervisor | NVIDIA, Israel | 2021 – Present

- Managed multi-site security operations, leading teams of officers and ensuring 100% operational compliance.
- Resolved complex operational challenges through effective problem-solving and cross-departmental communication.

Company Sergeant Major (Res.) | IDF

- Managing logistics for 150 soldiers and maintaining operational readiness under intense pressure.
- Leading specialized sniper teams and coordinating high-stakes field operations.

EDUCATION : B.Sc. in Computer Engineering | Ruppin Academic Center | 2021 – 2025

- **GPA: 87**. |**Dean's List:** 2022/2023, 2023/2024, 2024/2025.
- **Relevant Coursework:** Algorithms (95), Data Structures (100), Digital Signal Processing (91), VHDL Design (100), Operating Systems (86).